ANKARA UNIVERSITY
COMPUTER ENGINEERING DEPARTMENT
COM2553 File Organization

2024-2025 Fall Semester
Programming Assignment (PA) 2

Submission
This work is announced on December 4. Students should complete the assignment in two weeks and demonstrate their working programs after the lecture finishes on December 25. **You are expected to complete your work before (NOT during) the demonstration hours. You can submit your codes on e-Kampus until December 25, at 1:30 pm.**

You will implement a simple library catalog system in which a user can add a book record, delete a book record and search for a book with respect to ISBN Number. Each record has three fields:

- Book Title
- Author Name
- ISBN Number

Task a):
In the first part of the project, you assume that variable-length record size is used. Use a length indicator in the beginning of each record.

Your program must have a menu which is shown below.
Menu

1. Insert Record: Allows user to enter a new record to be stored. You may assume user enters only valid values. You are **NOT** required to check if the record already exists.
2. Print File: Prints all info in the file "books_variable.dat" in a neat format.
3. Find Record: Allows user to enter a valid ISBN Number to be searched and prints either the information about that book or a warning message for an unsuccessful search.

Task b):
In the second part of the project, you assume that fixed-length record size is used.

Your program must have a menu which is shown below.
Menu

4. Insert Record: Allows user to enter a new record to be stored. You may assume user enters only valid values. You are **NOT** required to check if the record already exists.
5. Print File: Prints all info in the file "books_fixed.dat" in a neat format.
6. Find Record: Allows user to enter a valid ISBN Number to be searched and prints either the information about that book or a warning message for an unsuccessful search.
7. Delete Record: Allows user to enter a valid ISBN Number to be deleted.

Make sure that you consider the following situations in your design:

- When you add a record, first look at the AVAIL LIST, then write the record. If there is a record available in the AVAIL LIST, write the record to a record AVAIL LIST points and make appropriate changes on the AVAIL LIST.
- When you delete a record, do not physically delete the record from file, just put a marker on the file and make appropriate changes on AVAIL LIST.
- The file should be compact after exiting the program, i.e. there should be no records marked as deleted in the file.

Regulations

You can work alone or in pairs. All the cheaters will be punished according to the Ankara University laws. Due date and time is important. Late submissions will not be accepted in any case.